Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/27/2020
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Flammability Limit Experiment (FLEX) - Phase II		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	858 534-5492
Organization Name:	University of California, San Diego		
PI Address 1:	Center for Energy Research		
PI Address 2:	Department of Mechanical and Aerospace Engineering		
PI Web Page:			
City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/12/2016	**End Date:**	09/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Hicks, Michael C.	**Contact Phone:**	216-433-6576
Contact Email:	Michael.C.Hicks@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2019 per NSSC information (Ed., 5/14/18) NOTE: End date changed to 1/31/2018 per M. Hicks/GRC and NSSC information (Ed., 9/26/17)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dryer, Frederick Ph.D. (Experiment Co-I: Princeton University) Choi, Mun Young Ph.D. (Experiment Co-I: University of Connecticut) Shaw, Benjamin Ph.D. (Experiment Co-I: University of California, Davis) Avedisian, C. Thomas Ph.D. (Experiment Co-I: Cornell University) Farouk, Tanvir Ph.D. (Experiment Co-I: University of South Carolina) Hicks, Michael M.S.M.E. (Experiment Co-I: NASA Glenn Research Center) Dietrich, Daniel Ph.D. (Experiment Co-I: NASA Glenn Research Center) Nayagam, Vedha Ph.D. (Experiment Co-I: Case Western Reserve University)		
Grant/Contract No.:	NNX16AR09A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	[Ed. note October 2016: Continues "Flammability Limit Experiment (FLEX)," grant NNX09AM33A] The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. The first phase of FLEX was a focused set of science objectives designed to address spacecraft fire safety concerns. The primary objectives were the assessment of flammability limits in a range of habitable atmospheres and fire suppressant efficacies. The second phase is FLEX-2 with a more expansive set of science objectives designed to investigate fundamental phenomena associated with (i) fuel sooting, (ii) development of reference fuels for fuel surrogates, (iii) low flow convective effects on diffusion flames, (iv) spacecraft fire safety, (v) droplet interactions in spray combustion, and (vi) condensed phase effects on burning using bi-component fuel blends. The research team for FLEX comprises externally funded Principal Investigators (PIs) and internally funded PIs. The lead researcher is Forman Williams for both FLEX and FLEX-2 and can be considered the de facto “PI” for FLEX (i.e., the combined experiments). The externally funded PIs are grantees affiliated with universities. The internally funded PIs are Michael C. Hicks (NASA-Glenn Research Center-GRC), Daniel L. Dietrich (NASA-GRC), and Vedha Nayagam (Case Western Reserve University). Flammability Limit Experiment (FLEX) Experiment PI: Forman Williams, Ph.D., University of California San Diego. FLEX and FLEX-2 Experiment External CoInvestigators (CoIs): Frederick Dryer, Ph.D., Princeton University. Task title: "Scientific Support for a Space Based Droplet Burning Experiment" Mun Young Choi, Ph.D., University of Connecticut. Task title: "FLEX Droplet Combustion Experiments in Microgravity" FLEX-2 Experiment External CoInvestigators (CoIs): C. Thomas Avedisian, Ph.D., Cornell University. Task title: "Combustion of Surrogate Fuel Droplets in Microgravity" Benjamin Shaw, Ph.D., University of California Davis. Task title: "Scientific Support for the FLEX-2 ISS (International Space Station) Experiment" Tanvir Farouk, Ph.D., University of South Carolina. Task title: "Mathematical Modeling and Simulation of Space Based Droplet Combustion – Scientific Support to FLEX and FLEX-2 Program" FLEX and FLEX-2 Experiment Internal CoInvestigators (CoIs): Michael C. Hicks, MSME, NASA Glenn Research Center. Daniel L. Dietrich, Ph.D., NASA Glenn Research Center. Vedha Nayagam, Ph.D., Case Western Reserve University.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research benefits life on Earth by improving knowledge of combustion processes needed by humankind.
Task Progress:	This research reflects an added scope of the FLEX project. The overall objective of the FLEX experiments (Flammability Limit Experiment) was to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives were to investigate influences of low gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. Experiments with hepane as the fuel and with methanol as the fuel were performed aboard the International Space Station (ISS). The project, in general, involved basic flammability-limit investigations relevant to fire safety issues for manned space flight. In standard fire-suppression techniques, a suppressant jet is directed towards a fire or a smoldering region. In considering these techniques for space applications, it is necessary to analyze the effect of suppressant transport to both the near-field and far-field of the region of interest. Fire prevention can also be implemented by modifying the atmospheric composition of the environment so as to reduce the likelihood of a self-sustaining combustion event. Finally, the flammability properties of materials can also be modified by the addition of materials which suppress ignitability, enhance charring of surfaces, or chemically inhibit self-sustained burning. The research addressed these aspects of fire safety in space. Some general directions of thrusts were as follows: 1. Map the flammability boundaries for liquid fuel combustion in reduced gravity. 2. Quantify the suppressant effectiveness of various gaseous suppressants over the range of candidate atmospheric pressures and O2 concentrations. 3. Develop predictive theoretical analyses and chemical-kinetic schemes to model flammability boundaries as a function effective gravitational acceleration on the unique ambient conditions encountered in space-exploration applications. 4. Devise improved and validated reduced (simplified) sub-models of important physical processes (chemical kinetics, etc.) that can be used in simulations of large scale, ‘realistic’ fires. 5. Determine combustion characteristics of liquid-fuel droplets of ethanol, decane, and liquid-fuel mixtures in varying atmospheres that can improve understanding of means for achieving clean and efficient power production. 6. Analyze cool-flame quasi-steady combustion of alkane droplets, especially heptane, but also decane and octane. 7. Develop theoretical descriptions of combustion and flame extinction in droplet burning, with special emphasis on cool-flame phenomena.

	8. Study the differences in cool-flame behavior for dodecane and lower alkanes, and offer possible explanations for the differences. 9. Complete asymptotic analyses of combustion of normal alkanes supported by cool-flame chemistry.
Bibliography Type:	Description: (Last Updated: 03/07/2024)
Articles in Peer-reviewed Journals	Williams FA, Nayagam V. "Cool-flame dodecane-droplet extinction diameters." Combusion and Flame. 2020 Feb;212:242-4. https://doi.org/10.1016/j.combustflame.2019.10.036 , Feb-2020
Articles in Peer-reviewed Journals	Williams FA, Nayagam V. "Quasi-steady combustion of normal-alkane droplets supported by Cool-Flame chemistry near diffusive extinction." Combustion Theory and Modeling. 2019;23(4):748-70. https://doi.org/10.1080/13647830.2019.1588380 , Mar-2019
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Williams FA. "Radiative extinction of burner-supported spherical diffusion flames: A scaling analysis." Combustion and Flame. 2019 Jul;205:368-70. https://doi.org/10.1016/j.combustflame.2019.04.027 , Jul-2019
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Williams FA. "Unsteady droplet combustion with fuel thermal expansion." Combustion and Flame. 2018 Sep; 195:216-9. https://doi.org/10.1016/j.combustflame.2018.01.035 , Sep-2018
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Hicks MC, Williams FA. "Radiative extinction of large n-alkane droplets in oxygen-inert mixtures in microgravity." Combustion and Flame. 2018 Aug;194:107-14. https://doi.org/10.1016/j.combustflame.2018.04.023 , Aug-2018
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Williams FA. "A Burke-Schumann analysis of dual-flame structure supported by a burning droplet." International Communications in Heat and Mass Transfer. 2017 Oct;87:84-9. https://doi.org/10.1016/j.icheatmasstransfer.2017.06.016 , Oct-2017
Articles in Peer-reviewed Journals	Nayagam V, Dietrich DL, Williams FA. "A Burke–Schumann analysis of diffusion-flame structures supported by a burning droplet." Combustion Theory and Modelling. 2017;21(4):646-57. https://doi.org/10.1080/13647830.2017.1280182 , Feb-2017